

Understanding cycling route choice behaviour through street-level images and discrete choice models

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Abstract

This study examines cyclists' preferences with respect to infrastructural and environmental attributes, including lane type, pavement quality, lane width, and the presence of greenery, water, and buildings. To do so, we conduct a novel stated route choice experiment where respondents evaluated alternatives varying in cycling environment, travel time and the number of traffic lights. For the survey design, we use thousands of real-world street-level images to represent the cycling environments, providing a more realistic choice context to respondents than what could have been achieved using a traditional stated choice experiment comprising tabular data with text and numbers. We analyse these data using two choice modelling approaches: a traditional discrete choice model with interpretable environmental attributes obtained through GIS data, and a recently proposed computer vision-enriched discrete choice model, which integrates computer vision into discrete choice models. Results show that cycling environments are important factors in explaining cyclist route choice behaviour, with cyclists preferring green areas and separated cycling lanes over urban areas and mixed-traffic roads. In addition, we use the developed models to map where the cycling network in Rotterdam, our case study, aligns—or conflicts—with cyclists' preferences, and to simulate targeted design changes to quantify their effects on predicted utility and route-choice probabilities. Model application results demonstrate its ability to provide planners with ex-ante evidence on the prospective effectiveness of proposed infrastructure upgrades.

Keywords: discrete choice models, computer vision, street-level imagery, stated choice experiment, cycling route choice, cycling environment

1 Introduction

Cycling has become a central element of sustainable transport strategies. It is widely promoted because of its benefits for health, accessibility, sustainability, and liveability (Pucher & Buehler, 2010). It is also recognised as a key measure for achieving the 2015 Paris Climate Agreement targets, including the goal of a 40% reduction in CO₂ emissions by 2030 (European Union, 2023; I&W, 2020). A crucial way for governments to encourage cycling is by providing high-quality infrastructure and supportive environments. Well-designed cycling environments have been shown to shift short-distance trips from car to bicycle (Fosgerau et al., 2023), improve perceived safety (Guo et al., 2023), and reduce accident risk (Costa, Azevedo, et al., 2024). To design such environments effectively, policymakers need a clear understanding of cyclists' preferences: how different attributes of cycling routes—such as surface quality, roadside greenery, and travel time—are valued, and what trade-offs cyclists are willing to make among them.

Discrete Choice Models (DCMs) are the dominant framework for investigating cyclists' preferences. By estimating utility functions from choice data, they quantify the relative importance of competing attributes in choice behaviour (McFadden, 1974). Both Stated Choice (SC) data, where hypothetical cycling route scenarios are presented to respondents (e.g., (Bovy & Bradley, 1985; Hardingham & Papantoniou, 2020; Liu et al., 2020; Majumdar & Mitra, 2018; Rossetti et al., 2018; Sener et al., 2009; Significance, 2022; Stinson & Bhat, 2003; Tarkkala et al., 2024; Uijtendewilligen et al., 2024; Vedel et al., 2017)), and Revealed Preference (RP) data, which comprise observed real-world cycling routes choices (e.g., (Bernardi et al., 2018; Broach et al., 2012; Skov-Petersen et al.,

2018; Ton et al., 2017)) have extensively been used. From these studies, it is evident that several characteristics significantly influence cyclists' behaviour. These include: (1) Cycling infrastructure, such as lane type, pavement type, road width, and road colour; (2) Built environment factors, like greenery, buildings, street lights, parking and land-use; (3) Traffic-related attributes, including car traffic intensity, speed limits, cycling intensity and the presence of buses; and (4) Route-specific attributes, including travel time, travel distance, number of turns, intersections, and traffic lights. Notably, many studies find that the presence of dedicated cycling infrastructure and travel time are among the most influential factors to cyclists' route choices (Bovy & Bradley, 1985; Broach et al., 2012; Hardinghaus & Papantoniou, 2020; Rossetti et al., 2018; Sener et al., 2009; Skov-Petersen et al., 2018; Stinson & Bhat, 2003; Tarkkala et al., 2024; Tilahun et al., 2007; Ton et al., 2017; Vedel et al., 2017).

The majority of SC studies rely on textual descriptions to convey attributes of the environment, such as “the width of the bicycle path”, “amount of street greenery”, or perceptual labels, such as “busy”, and “safe” (e.g. (Bovy & Bradley, 1985; Liu et al., 2020; Majumdar & Mitra, 2018; Sener et al., 2009; Stinson & Bhat, 2003)). Several studies incorporate visuals, such as sketches, abstract illustrations, or manually edited images, typically using a small, fixed set of carefully designed illustrations presented uniformly across respondents (Hardinghaus & Papantoniou, 2020; Rossetti et al., 2018; Significance, 2022; Tarkkala et al., 2024; Uijtdewilligen et al., 2024; Vedel et al., 2017). While these studies aim to represent the cycling environment in line with the researcher's intentions, they are limited in several significant ways. First, simplified visuals and textual descriptions often cannot accurately represent the full complexity of real-world environments (Florio et al., 2023). While these representations aim to capture key elements such as bike infrastructure or traffic, they often omit important contextual features like surface texture, vegetation, and the overall sense of safety or aesthetics (Florio et al., 2023; Hevia-Koch & Klinge Jacobsen, 2019). Moreover, when such features are visually present, they are seldom incorporated as explicit parameters in the choice models (See Rossetti et al. (2018) and Tarkkala et al. (2024), in which trees appear in the visualisations yet no corresponding “tree presence” coefficient is estimated). Simplified visuals and textual descriptions also place a cognitive burden on respondents, requiring them to mentally reconstruct the environment, which may result in misinterpretation and biased preference estimates (Childers et al., 1985; Pinker, 1988). Finally, the use of a limited set of images can make results sensitive to specific images shown, thereby undermining external validity and generalisability (Hevia-Koch & Klinge Jacobsen, 2019). Consequently, the understanding of cyclists' preferences for cycling environments remains incomplete, largely due to the way these environments are presented in stated choice experiments.

Unlike text descriptions or abstract visualisations, street-level imagery captures much of the complexity of cycling environments and allows researchers to present more ecologically valid contexts. Moreover, the growing availability of such imagery, together with advances in computer vision, enables analysis at scale (Kirillov et al., 2023). Recognising this potential, a parallel strand of research has emerged that examines cyclists' *perceptions* of the environment using street-level images. Typically, their workflow is as follows: respondents are shown paired street-view images and asked which environment feels safer or more pleasant for cycling; the responses are then used to train machine learning models that assign perceptual scores to new unseen images (Dubey et al., 2016). The resulting cycling environment utilities are then mapped to reveal spatial patterns of cycling perception (Costa, Marques, et al., 2024; Ito & Biljecki, 2021). Moreover, using pixel-level segmentation (e.g., percentages of greenery, buildings, or traffic elements), they are able to identify which visual features drive these cycling perceptions (Zeng et al., 2024). A comprehensive overview of this line of research is provided by Zhang et al. (2024) and Biljecki and Ito (2021). However, although perceptions provide valuable insights, they cannot explain cyclists' route choice behaviour. Perceptions are subjective interpretations of the environment (Wade & Swanston, 2013). Unlike preferences (which are formally rooted in choice theory (Lancaster, 1966; Luce, 1959; Samuelson, 1938)), perceptions do not capture the trade-offs individuals are willing to make that ultimately drive cycling choices.

In sum, the cycling literature splits into two strands with complementary strengths and limitations. The discrete-choice strand quantifies cyclists' trade-offs, but typically represents environments with text or stylised visuals, which constrains validity. The perception strand exploits street-level imagery at scale, but elicits subjective perceptions rather than preferences, and therefore does not identify the trade-offs that govern route choice behaviour. As such, current literature provides limited guidance for reliably designing and validating cycling infrastructure that aligns with cyclists' preferences.

This study addresses this gap by eliciting cyclists' preferences through a SC experiment using real street-level images. In our experiment, respondents chose between route alternatives that varied in the visual environment (image), travel time, and the number of traffic lights. To construct the experiment, we assembled a large set of cyclist-perspective images from the Municipality of Rotterdam's cycling infrastructure database. We then estimated

three models: (i) a conventional DCM which ignores the images (as a benchmark), (ii) a conventional DCM with explicitly coded environmental attributes, and (iii) a computer-vision-enriched DCM (CV-DCM) that learns salient visual features end-to-end from the imagery (van Cranenburgh & Garrido-Valenzuela, 2025). We apply the latter model to map where Rotterdam's cycling network aligns—or conflicts—with cyclists' preferences, and we simulate targeted design changes to quantify their effects on predicted utility and route-choice probabilities, providing planners with ex-ante evidence on the prospective effectiveness of proposed upgrades.

The remainder of this article is organised as follows. Section 2 describes our methods: data collection, the design of the stated-choice experiment, and the model specifications, including the conventional discrete-choice models and the computer-vision-enriched variant. Section 3 presents the estimation results and the model application to our case study: the cycling network in Rotterdam, the Netherlands. Finally, Section 4 provides a conclusion and discusses avenues for further research.

2 Methods

This Section covers three subparagraphs. First, we detail the collection of the images and cycling environment attributes in Paragraph 2.1. Second, we describe the design of the stated-choice experiment in Paragraph 2.2. Third, we present the estimated models in Paragraph 2.3.

2.1 Data collection

In the following we outline the process of data collection in two parts. First the the collection and categorisation of street-level images for the SC experiment. Then the construction of a tabular dataset of cycling environment attributes for modelling preferences.

2.1.1 Street-level image collection and analysis

We use street-level images in our SC experiment to represent the cycling environment. Accordingly, our data collection process begins with gathering street-level images taken from the cyclist perspective. These are obtained from the municipality of Rotterdam's image database (Cyclomedia, 2023) and captured between late February and early April 2023, reflecting typical late winter conditions. The full image database contains approximately 600,000 instances, including images from motorways, intersections, and areas not accessible to cyclists. We restrict image sampling to network segments where cycling is suitable. Additionally, we collect information about the key attributes in each image to support meaningful trade-off analyses and achieve a diverse representation of cycling environments. To do so, we first retrieve the cycling network (Fietzersbond, 2020), which consists of segments defined as stretches between intersections. For each segment, we place a midpoint and retrieve the nearest image within a 4-metre radius, ensuring that only locations suitable for cycling are included. Segments shorter than 3 metres (typically intersections) are excluded, as are images located in pedestrian zones, on bridges, or in parking areas. This process yields approximately 13,000 panoramic 360-degree images. Subsequently, we convert the panoramas into 90-degree forward-facing images aligned with the cycling direction.

Next, we analyse and categorise the images according to three key attributes: cycling infrastructure, built environment, and speed limit. These attributes are selected based on findings from previous cycling route choice studies. We link spatial datasets to the images to assign attribute levels to each image. Table 1 presents the attributes and their corresponding levels. We categorise the images into the 15 most frequently occurring attribute combinations. Categories with very few images are merged with similar ones to avoid over-representation of rare cases. These 15 categories serve as the image attribute levels in the SC design, providing a meaningful trade-off analysis as well as a balanced representation of cycling environments.

Table 1: Characteristics and corresponding levels

Cycling Infrastructure <i>Fietzersbond (2020)</i>	Built Environment <i>BGT (2024)</i>	Speed Limit <i>Rijkswaterstaat (2024)</i>
Solitary cycling lane	Recreational area	Not applicable
Separated cycling lane	Residential area	30 km/h
Mixed-traffic road	Neighbourhood access	50 km/h
Cycling suggestion lane	Main road	
	Industrial area	

2.1.2 Cycling environment attributes

To estimate marginal utilities for cycling environment attributes visible in the street-level images, we enriched the image sample with structured attribute data. We considered two approaches. First, a computer-vision segmentation pipeline (e.g., Mask2Former) could extract pixel-level classes (trees, buildings, cars, roads, etc.). However, such pipelines are constrained by a predefined label set and struggle to recover several quantities central to cyclists' route choices (e.g., lane width, surface condition). Second, we could fuse external GIS sources that describe land use, cycling infrastructure and traffic conditions. Given the Municipality of Rotterdam's detailed GIS datasets, the latter approach offers greater attribute richness and measurement accuracy. We therefore adopted the GIS-fusion route.

Concretely, we spatially joined each geolocated image to multiple GIS layers to derive: (1) *cycling-infrastructure* features (lane type, pavement, lane width); (2) *built-environment* features (trees, grass, water, industrial and other buildings); and (3) *traffic and environmental* features (posted car speeds, weather and brightness conditions). Timestamps associated with the images enabled alignment to historical weather records from Visual Crossing (2024). Different spatial buffers were applied during the merge: large features (e.g., buildings, water) were included if present within 20 metres of the image location, while smaller features (e.g., grass, planting, parking, tram lines) used a 10 metre buffer. Trees were counted along the length of the corresponding cycling segment. The resulting database comprises approximately 13,000 cyclist-perspective images, each linked to the cycling-environment attributes used for (i) sampling images for the stated-choice experiment and (ii) subsequent model estimation. Table 2 lists the full attribute set and provides example values for three images; the corresponding images are shown in Figure 1.

Table 2: Collected attributes, linking method, and values for three example images

Attribute	Levels	Source	Linking method	Example images		
				IMG1	IMG2	IMG3
Built Environment						
Trees	Count	BGT	Count within 20 m buffer	0	2	4
Grass	Yes/No	BGT	Present within 10 m	No	Yes	Yes
Plants	Yes/No	BGT	Present within 10 m	No	Yes	Yes
Water	Yes/No	BGT	Present within 20 m	No	Yes	No
Buildings	Yes/No	BGT	Present within 20 m	Yes	Yes	No
Industrial buildings	Yes/No	BGT	Present within 20 m	No	No	No
Cycling Infrastructure						
Infrastructure type	Categories	Fietsersbond	Initial network layer	Mixed-traffic road	Suggestion lane	Separated lane
Pavement type	Asphalt/Clinkers	Fietsersbond	Initial network layer	Asphalt	Asphalt	Asphalt
Parking	Yes/No	BGT	Present within 10 m	Yes	No	No
Width	Metres	BGT	Present within 10 m	5.0	4.7	2.5
Colour	Red/Not red	BGT	Present within 10 m	Grey	Red	Red
Tram line	Yes/No	BGT	Present within 10 m	Yes	Yes	No
Traffic and Environment						
Car speed	30/50 km h ^{−1}	Rijkswaterstaat	Present within 10 m	50	50	N/A
Weather	Sunny/Cloudy	Visual Crossing	Image timestamp	Cloudy	Cloudy	Sunny
Brightness	[0, 255]	Calculated	Computed from image	115	108	153



(a) IMG1



(b) IMG2



(c) IMG3

Figure 1: Example images

2.2 Stated choice experiment

This section outlines the setup of the SC experiment, including the selection of attributes and their levels, the experimental design and the layout and implementation of the survey. It also includes a descriptive analysis of the respondents' sample.

2.2.1 Selected attributes and ranges

In our SC experiment, respondents are presented with two hypothetical cycling routes. We conceptualise the cycling route alternatives using three attributes: the cycling environment (presented using a street-level image), the travel time and the number of traffic lights. These attributes are selected due to their consistent importance in cycling route choice behaviour and their relevance for policy and design interventions. Travel time has repeatedly been identified as a dominant factor in route choice decisions (Bovy & Bradley, 1985; Broach et al., 2012; Significance, 2022; Skov-Petersen et al., 2018; Stinson & Bhat, 2003; Ton et al., 2017). The number of traffic lights reflects interruptions to routes and has been shown to negatively affect cyclists' preferences (Broach et al., 2012; Sener et al., 2009; Skov-Petersen et al., 2018; Tarkkala et al., 2024; Ton et al., 2017; Uijtdewilligen et al., 2024; Vedel et al., 2017). While car speed and traffic intensity have also been found important in previous studies, they are not included as separate attributes in the design. As these factors are already partially conveyed through the images, including them would potentially create data congruency issues, see (Siffringer & Alahi, 2023). We use 8, 11, and 14 minutes for the travel time attribute, representing realistic durations for urban cycling trips; for traffic lights, we use 1, 2, and 3. The image attribute comprises 15 levels corresponding to the cycling environment categories defined in Section 2.1.1.

2.2.2 Efficient design

We employ an efficient design to maximise the information obtained from the SC experiment, resulting in more reliable parameter estimates for a given number of observations (Rose & Bliemer, 2009). Prior estimates for the model parameters are based on findings from Significance (Significance, 2022). The design is created using Ngene software (ChoiceMetrics, 2024), generating two blocks of 15 choice tasks. The design includes labelled attributes, namely travel time, number of traffic lights, and the image category, with the latter reflecting a combination of infrastructure type, speed limit, and built environment type. In the experiment, for a given combination of infrastructure type, speed limit, and built environment type, images were randomly drawn from the image database.

2.2.3 Survey design

The survey was implemented on a web platform developed using the Python package Dash (Plotly, 2024). The survey comprises three parts: 1) the SC experiment; 2) Perception questions involving the rating of images on traffic safety, social safety, and aesthetics, and 3) Questions about socio-demographic characteristics and cycling behaviour. Figure 2 shows an example of a choice task from the experiment. Respondents are given the following instructions:

- Imagine you are cycling from your work, train station, school, or daily activity to your home.
- There are two cycling routes you can take.
- The routes differ only in **travel time**, the number of **traffic lights**, and the **cycling environment** shown in the photo.
- In all other aspects, such as the weather, the cycling routes are the same.
- **Which route would you choose?**
- You can assume the following:
 - You are cycling alone.
 - You are not in a hurry.
 - The photo gives a good idea of what the **entire route** looks like.
 - The travel time is the **total time** for the route, **including waiting at traffic lights**.
 - The weather is partly cloudy with no rain.

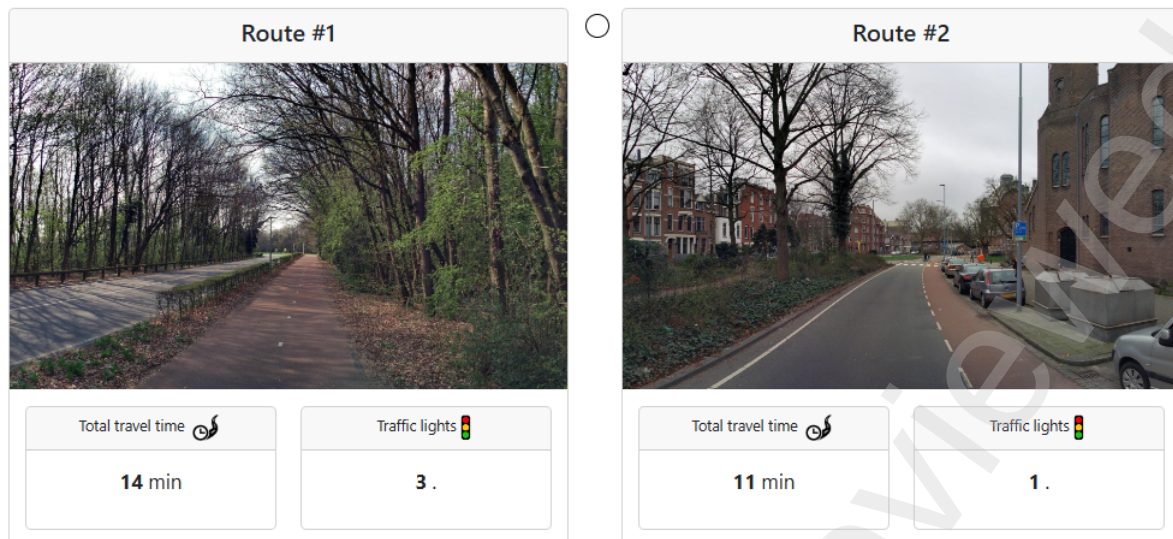


Figure 2: Example of a choice task

2.2.4 Survey implementation

The survey was conducted between late September and early October 2024. Respondents are recruited through a panel company and are financially compensated Cint (2024). The respondents are aged 18 years and above who regularly engage in cycling. To ensure that the sample is broadly representative of the Dutch population aged 18 and above, respondent entry quotas are applied based on gender, age group, and region, using distribution targets taken from national statistics (CBS, 2024a). Quotas were enforced at screening: once a gender–age–region group reached its target, additional candidates in that group could not proceed. Respondents who completed the choice tasks too quickly are excluded to ensure adequate engagement with the choice tasks. In total, 753 valid responses were collected, corresponding to 11,295 observed choice tasks and involving 6,484 unique images.

Respondents were randomly assigned at the outset to either the training set (80%) or the test set (20%). Each set was linked to a disjoint pool of images, ensuring that no image shown to the training set appeared in the test set. This respondent-level, image-disjoint split prevents data leakage —i.e., models memorising specific images rather than learning generalisable visual attributes. Each respondent is presented with 15 choice tasks. The choice task order and the position of alternatives (left/right) are randomised to prevent potential ordering effects. To avoid repetition, respondents would not see the same image more than once.

2.2.5 Descriptive analysis

Respondents are asked about socio-demographic characteristics and their cycling habits. Overall, the sample is balanced in terms of age, gender, education, income and employment status and broadly reflects the national population. However, the sample over-represents individuals of Dutch origin (92%), which differs from the national population (72%) (CBS, 2024a). This may influence the generalisability of the results, particularly with regard to cycling behaviour and perceptions, which can vary across cultural backgrounds (Heinen et al., 2010). Approximately half of the participants identify cycling as their most frequent mode of transport. One-third of the respondents primarily cycle for commuting, while two-thirds cite other purposes such as recreation, shopping, or exercise. Previous research suggests that commuting cyclists tend to prioritise speed and directness, whereas recreational cyclists place more weight on the cycling environment (Bernardi et al., 2018). Respondents are also asked to rate the importance of the three route attributes in their decision-making. On average, the cycling environment (as presented by the image) is rated as the most important attribute (7.6), followed by travel time (6.1) and the number of traffic lights (5.5). While self-reported importance may not fully align with actual choice behaviour, these ratings offer a first insight into perceived priorities and suggest that the visual environment plays a significant role in route preferences, which is in line with other SC studies (van Cranenburgh & Garrido-Valenzuela, 2024).

2.3 Discrete choice models

Section 2.3.1 introduces the three models and Section 2.3.2 briefly revisits the computer vision-enriched discrete choice model extension and details the CV-DCM training procedure.

2.3.1 Model overview

To examine cyclists' preferences with respect to the cycling environment, we estimate three discrete choice models. Each model exploits the information concerning the cycling environment in a different way. All three models are grounded in Random Utility Maximisation (RUM) and are estimated as MNLs (i.e. they assume unobserved utilities are i.i.d. Extreme Value Type I distributed). The three models are described below:

1. **Model 1 (baseline).** A standard linear-additive RUM-MNL model using only travel time and the number of traffic lights; hence, the cycling environment is omitted. This model serves as a benchmark: improvements in fit achieved by models that incorporate cycling-environment information (coded as attributes or images) relative to this baseline shed light on the importance of the cycling environment for explaining the cyclists' route choice behaviour. Equation 1 shows its utility function.

$$U_{in} = \beta_{tt} \cdot tt_{in} + \beta_{tl} \cdot tl_{in} + \varepsilon_{in} \quad (1)$$

2. **Model 2 (GIS-coded cycling-environment attributes).** A standard linear-additive RUM-MNL that augments the baseline with the GIS-derived attributes listed in Section 2.1.2, alongside travel time and the number of traffic lights. Because these inputs are explicitly coded numerical attributes, the model yields interpretable marginal utilities for specific environmental features—e.g., trees, pavement type, lane width, and buildings—and thus quantifies their association with route choice. The indirect utility specification is given in Equation 2, where x_{imn}^{gis} denotes the interpretable cycling-environment attributes derived from GIS data.

$$U_{in} = \beta_{tt} \cdot tt_{in} + \beta_{tl} \cdot tl_{in} + \sum_m \beta_m \cdot x_{imn}^{gis} + \varepsilon_{in} \quad (2)$$

3. **Model 3 (CV-DCM).** A computer-vision-enriched discrete choice model that integrates an image encoder into the utility function to ingest street-level images directly, rather than relying on GIS-coded attributes (van Cranenburgh & Garrido-Valenzuela, 2025). Because it learns visual signals from the same images shown to respondents, it can be expected to deliver higher predictive accuracy compared to Model 2. However, this comes at a cost to interpretability, as the image-derived coefficients relate to latent embedding dimensions rather than semantically interpretable attributes. The indirect utility function of this model is given in Equation 3, where z_{ikn}^{img} represents the image embedding of the street-level image. Since this model is newer and requires training (as opposed to estimation), the next section revisits this model briefly.

$$U_{in} = \beta_{tt} \cdot tt_{in} + \beta_{tl} \cdot tl_{in} + \sum_k \beta_k \cdot z_{ikn}^{img} + \varepsilon_{in} \quad (3)$$

2.3.2 Computer vision-enriched discrete choice model

The computer-vision discrete-choice model augments the traditional discrete-choice framework with a computer-vision component. In brief, this model uses an image encoder to produce an embedding for each image. This embedding is a 1×1000 vector that contains the image's salient features in a compressed form (Touvron et al., 2021). The CV-DCM maps the embedding, z_{ikn} , linearly onto utility, alongside the usual numerical attributes, see the indirect utility function in Equation 3. Because the embedding dimensions are not semantically labelled (i.e. they need not correspond to objects such as “trees” or “lane type”), the associated coefficients are not directly interpretable. By contrast, the coefficients of the explicit numeric attributes—i.e. travel time and number of traffic lights—retain their standard behavioural meaning. Figure 3 summarises the overall architecture; further methodological detail is provided by van Cranenburgh and Garrido-Valenzuela (2025).

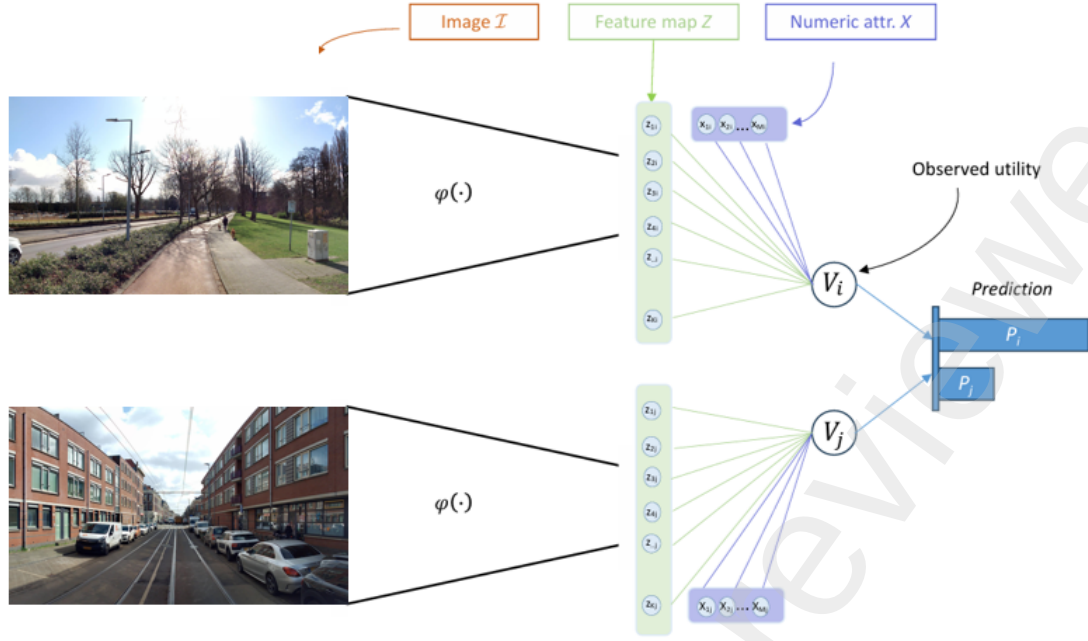


Figure 3: Model structure of the CV-DCM.
Adapted from van Cranenburgh and Garrido-Valenzuela, 2025

Furthermore, to train the CV-DCM, we minimise the cross-entropy loss, which is equivalent to maximising the log-likelihood under the RUM framework (Aouad & Désir, 2023). We add an L2 regularisation penalty γ to mitigate overfitting. This is only applied to the feature-extractor weights w that produce the image feature map. We do not regularise the preference parameters—neither the numeric attribute coefficients β_m nor the image feature coefficients β_k —to avoid introducing bias in the preference estimates. The full objective is given in Equation 4, where β denotes the preference parameters and w represents the computer vision weights.

$$w, \beta = \arg \min_{w, \beta} \left(\underbrace{\frac{1}{N} \sum_{n=1}^N \sum_{j=1}^J y_{nj} \log(P_{nj} | X_{nj}, S_{jn}, \beta)}_{\text{Cross-entropy loss}} + \underbrace{\gamma \sum_{r=1}^R w_r^2}_{\text{L2 regularisation}} \right) \quad (4)$$

We implement and train the CV-DCM using PyTorch. PyTorch is designed to build and train deep neural networks and facilitate GPU support, accelerating training. We conduct a hyperparameter search to identify a well-performing model configuration. In doing so, we systematically vary the optimisation algorithm, learning rate, batch size, and L2 regularisation strength. Table 3 summarises the hyperparameter combinations explored. The final model is selected based on the lowest cross-entropy loss on the test set, using early stopping with a patience of five epochs. The best-performing configuration employs stochastic gradient descent (SGD) with a learning rate of 1×10^{-5} , a batch size of 20, and an L2 weight-decay factor of 0.1, with the selected values highlighted in bold in Table 3.

Table 3: Hyperparameter configurations explored for CV-DCM training

Hyperparameter	Values Tested
Optimisation Algorithm	SGD , Adam
Learning Rate	1×10^{-5} , 1×10^{-4}
Batch Size	12, 16, 20 , 24
L2 Weight Decay (γ)	0, 0.05, 0.1

3 Results

This section comprises two parts. Section 3.1 reports the training and estimation results for the three models; Section 3.2 presents applications of our models to the city of Rotterdam, illustrating how the models' outputs can support planners with preference-based diagnostics of the cycling network.

3.1 Training and estimation results

Table 4 reports our estimation/training results. We can make the following observations based on Table 4. Model 1, which contains only travel time and the number of traffic lights, attains a very low model fit (ρ^2) of 0.036 on the test set and the highest cross-entropy loss of 0.668. By contrast, Model 2 (GIS-coded cycling-environment attributes) and Model 3 (CV-DCM) raise the model fit on the test set to 0.092 and 0.119, respectively. These results show that cyclists' route choices are strongly shaped by the cycling environment, that the GIS-based attributes in Model 2 capture behaviourally relevant information, and that the computer-vision component in Model 3 extracts meaningful information from street-level imagery to explain route choice behaviour. Finally, comparing training and test statistics reveals only a modest difference in model fit. For Model 2, the model fit drops from 0.099 on the training set to 0.092 on the test set; for Model 3, the corresponding decline is from 0.127 to 0.119. These small gaps indicate only very limited over-fitting and suggest that both models generalise well to unseen choice tasks. The CV-DCM comprises approximately 86 million trainable weights, which involves a training time of about 1.5 hours on a midrange GPU. Although this is longer than the estimation time of a standard discrete choice model, it remains well within practical limits.

In line with theoretical expectations, the CV-DCM (Model 3) yields the best generalisation performance as it is better able to extract more information, such as transient traffic, spatial openness, and continuity, from the images than Model 2, which uses GIS data sources for the environmental attributes. The CV-DCM's deep network architecture further accommodates nonlinearities and interactions among these attributes encoded in the image, enabling it to explain a larger share of the variance in cyclists' utilities derived from the cycling environment. On the other hand, the CV-DCM might overlook fine-scale or rare markings (e.g., tram rails or the narrow white line that denotes a cycling suggestion lane) that are explicitly encoded in the GIS variables.

The β coefficients represent marginal utilities and can be used to compute time-equivalent trade-offs. Specifically, the time-equivalent to reduce one traffic light can be determined by taking the ratio of the coefficients, β_{tl}/β_{tt} . Based on Model 2, eliminating one traffic light is valued at 1.38 minutes of travel time. The CV-DCM yields a very similar estimate of 1.48 minutes, indicating that this trade-off is robust to model specification.

Although the fit of Model 2 does not attain as good a fit as the CV-DCM, it provides more explanations because the betas associated with the cycling environment features are interpretable. Most coefficients of Model 2 in Table 4 retain the expected signs. The model reveals that cyclists strongly prefer separated cycling lanes over mixed-traffic roads. Specifically, the findings indicate that cyclists are willing to extend a trip of around 11 minutes by 1.5 minutes to use a separate cycling lane rather than a mixed-traffic road, as implied by the ratio $\beta_{separatedlane}/\beta_{tt}$. Similarly, natural features such as trees, water, and grass are significantly preferred, whereas (industrial) buildings, parking spaces, and trams are less favoured by cyclists. Industrial buildings generate the largest disutility ($\beta = -0.63$), plausibly reflecting their low aesthetic quality and the prevalence of trucks and other motorised vehicles. Additionally, cyclists exhibit a clear preference for asphalt surfaces over brick paving. Although respondents were instructed to ignore the weather in the experiment, the coefficient β_{sunny} is significant, indicating a clear preference for images captured in sunny conditions.

Several candidate variables, namely cycling suggestion lane, plants, and width are statistically insignificant at the 5 % level. Their lack of significance may come from indifference, but measurement issues could also play a role. For example, the plant variable covers only vegetation in public space, omitting greenery in front gardens; cycling suggestion lanes typically occur on busy roads which may not be fully captured by other GIS variables; many images depict quiet streets, which may reduce the importance of the width (Greibe & Buch, 2015); and width is sometimes inaccurately recorded in the GIS data.

Table 4: Training and estimation results

	Model 1			Model 2			Model 3		
Model type	RUM-MNL			RUM-MNL			CV-DCM		
Number of parameters	2			16			86m		
Estimation time	<1 sec			<1 sec			1.5 hour		
Train Set (N = 9,146)									
Log-Likelihood	-6,157			-5,704			-5,533		
ρ^2	0.028			0.099			0.127		
Cross-entropy	0.674			0.624			0.605		
Test Set (N = 2,128)									
Log-Likelihood	-1,421			-1,339			-1,300		
ρ^2	0.036			0.092			0.119		
Cross-entropy	0.668			0.629			0.611		
	est	s.e.	p-val	est	s.e.	p-val	est	s.e.	p-val
Parameters									
β_{tt}	-0.11	0.012	0.00	-0.23	0.0139	0.00	-0.18	0.0137	0.00
β_{tl}	-0.23	0.021	0.00	-0.31	0.0215	0.00	-0.27	0.0219	0.00
$\beta_{separatedlane}$				0.36	0.054	0.00			
$\beta_{suggestionlane}$				-0.05	0.054	0.37			
β_{bricks}				-0.11	0.041	0.01			
β_{width}				0.00	0.010	0.91			
$\beta_{parking}$				-0.09	0.039	0.02			
β_{tram}				-0.23	0.107	0.03			
β_{trees}				0.10	0.015	0.00			
β_{gras}				0.24	0.038	0.00			
β_{water}				0.18	0.038	0.00			
β_{plants}				0.06	0.038	0.12			
β_{house}				-0.31	0.049	0.00			
$\beta_{industrial}$				-0.63	0.059	0.00			
β_{sunny}				0.09	0.033	0.01			
$\beta_{brightness}$				0.04	0.008	0.00			
Traffic light [min]	2.12	0.2522	0.00	1.38	0.1034	0.00	1.48	0.3985	0.00

3.2 Model applications to Rotterdam

In this section, we apply our best-performing model—the CV-DCM—to illustrate how model outputs can support planners with preference-based diagnostics of the cycling network. Our case study is Rotterdam, the Netherlands' second-largest city, with approximately 660,000 residents (CBS, 2024b; Custers & Willems, 2024). The municipality aims to shift short trips from car to bicycle (Gemeente Rotterdam, 2019), improve perceived safety (Gemeente Rotterdam, 2023), and reduce traffic accidents (Fietsersbond, 2023). These objectives make Rotterdam a suitable setting for assessing how visual features of the cycling environment relate to cyclists' preferences.

We apply the trained CV-DCM to ~13,000 cyclist-perspective street-level images sampled across Rotterdam's cycling network (Section 2.1.1 details the sampling procedure). For each image, the model returns a systematic utility value, which we interpret as a preference-consistent indicator of the local cycling environment (higher values imply a greater predicted propensity to choose routes featuring such environments, *ceteris paribus*). We use these utilities to:

- Map the spatial distribution of the cycling environment utility, highlighting where the cycling network best and worst aligns with cyclists' preferences (Section 3.2.1);
- Identify the most salient visual attributes associated with the highest and lowest cycling environment utilities to highlight the recurring cues that distinguish preferred from non-preferred cycling environments. (Section 3.2.2);
- Examine associations between infrastructure types and built environment types and cycling environment utility (Section 3.2.3);
- Simulate design changes to provide planners with ex-ante evidence on the prospective effectiveness of proposed infrastructure upgrades. (Section 3.2.4).

3.2.1 Spatial distribution of the cycling environment utility

To reveal where Rotterdam's network best and worst aligns with cyclists' preferences, we map the cycling environment utility across all links of the cycling network. Because these utilities are relative, the colour scale highlights areas that outperform or underperform compared with the rest of the city. We then classify the utilities into five clusters with Jenks natural breaks, which minimise variance within clusters and maximise variance between clusters (Chen et al., 2013), see legend. Figure 4 shows the resulting spatial distribution of the cycling environment utility for the cycling network of Rotterdam.

It is notable that the centre of Rotterdam is coloured red and orange (underperforming), while the surrounding regions are coloured yellow and light green (average to above average). The recreational areas and an area to the north-west, which is mostly composed of meadows, are represented by a dark green colour (overperforming). This contrast suggests that cyclists prefer green spaces over densely populated urban areas. A similar pattern is reported by Costa, Marques, et al. (2024) who found that cyclists perceive less urbanised areas with high levels of vegetation as more traffic safe.

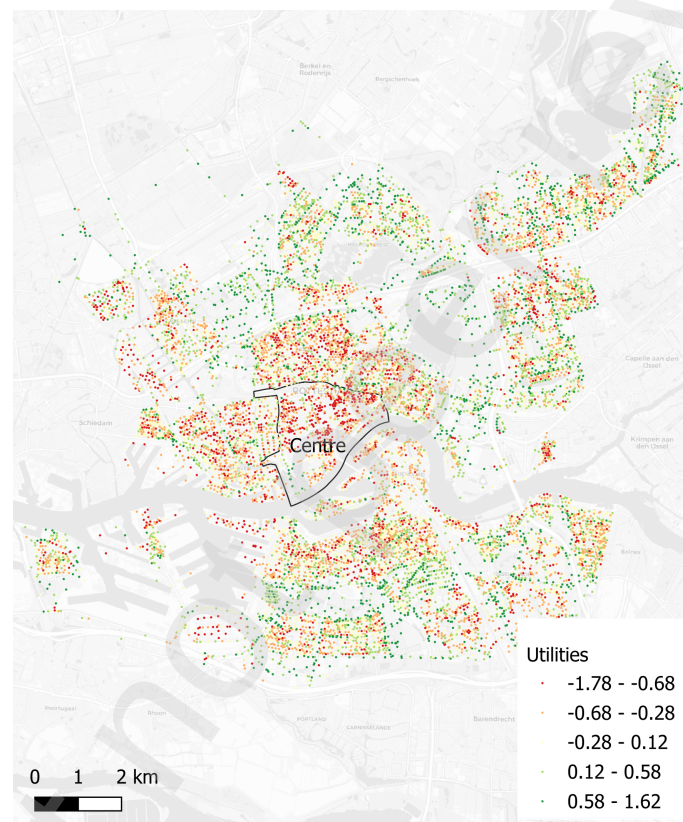


Figure 4: Cycling environment utility for the cycling network of Rotterdam

3.2.2 Associations between cycling environment utility and visual attributes

To further understand what drives these cycling environment utilities, we explore the visual attributes that are associated with them. Specifically, we examine the visual attributes that associate with the highest and lowest cycling environment utilities. Accordingly, Figure 5 shows nine random images from the highest and lowest utility clusters used to produce Figure 4. We observe that the highest cycling environment utilities are associated with separated cycling lanes, green and open surroundings, and few buildings or traffic. In contrast, the lowest cycling environment utilities are associated with images of urban areas featuring buildings, vehicles, and narrow mixed-traffic roads. This pattern aligns with prior findings that cyclists prefer routes with greenery and separated lanes (Rossetti et al., 2018; Tarkkala et al., 2024; Uijtdewilligen et al., 2024).



(a) Images from highest cluster ($0.58 < V^{img} < 1.62$).

(b) Images from lowest cluster ($-1.78 < V^{img} < -0.68$).

Figure 5: Images sampled from the highest (left) and lowest (right) clusters based on cycling environment utilities

3.2.3 Associations between cycling environment utility and infrastructure and built environment types

To further examine cyclists' preferences across cycling infrastructure types, we group the images based on the type of cycling lane and the maximum allowed car speed on the road. Figure 6 presents the associations between cycling environment utility and cycling infrastructure (including speed limit) in a boxplot, ordered by increasing mean utility. Utilities are substantially higher for separated cycling lanes, whereas the difference between mixed-traffic roads and cycling suggestion lanes is marginal. This further supports the notion that the physical separation of the cycling lane from traffic is important to cyclists. The maximum car speed (30 km/h or 50 km/h) appears to be weakly associated with cycling environment utility. In contrast, previous studies have shown that higher car speed significantly reduce the attractiveness of a cycling route (Hardinghaus & Papantoniou, 2020; Rossetti et al., 2018; Sener et al., 2009; Significance, 2022; Vedel et al., 2017). A possible explanation is that respondents struggled to infer traffic speeds from the presented images.

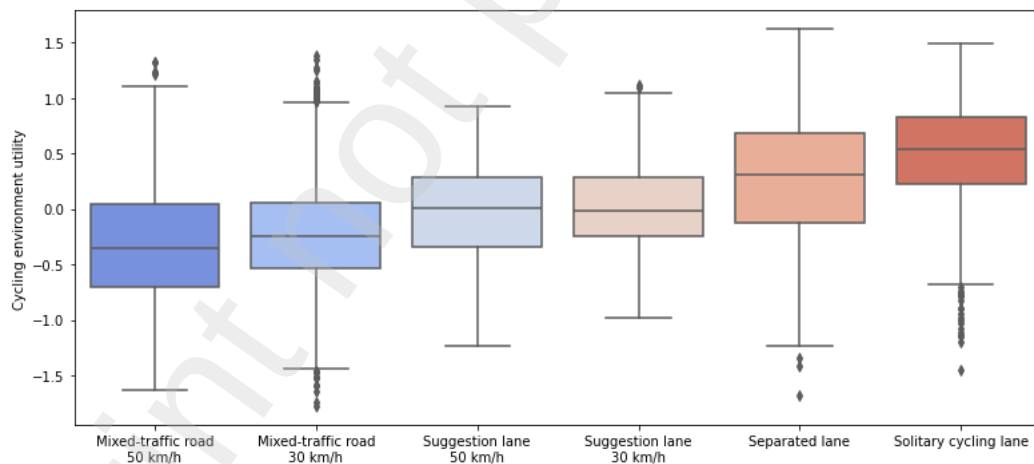


Figure 6: Cycling environment utility by cycling infrastructure and car speed

In addition, we examine how different built environment types are associated with cycling environment utility, see Figure 7. The importance of built environment type to cycling route choice behaviour has rarely been studied; to the best of the authors' knowledge, only one study has examined this explicitly (Tarkkala et al., 2024). Indeed, most research relies on greenery indicators to represent the built environment. Figure 7 shows that recreational areas, which characteristically combine separated cycling lanes with greenery, are associated with the highest utilities. These are followed by main roads and neighbourhood access roads. Residential and industrial areas are associated with the lowest utilities. Our results align with Tarkkala et al. (2024), who report higher disutility for local streets than for main roads, and with earlier evidence that cyclists strongly favour green surroundings (Hardinghaus & Papantoniou, 2020; Liu et al., 2020; Vedel et al., 2017).

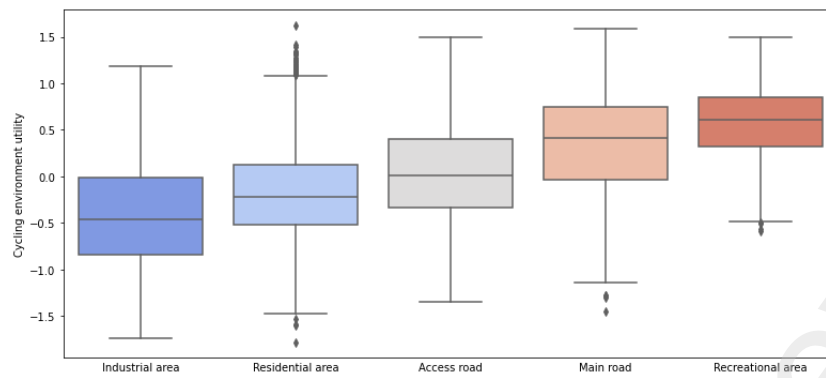


Figure 7: Cycling environment utility by built environment type

3.2.4 Ex-ante evaluation of cycling environment interventions

The municipality of Rotterdam has earmarked several roads for upgrades, including the road shown in the centre of Figure 8. To examine how specific cycling environment design interventions affect cyclists' utility derived from the cycling environment in a route choice situation, we generated four synthetic variants of the baseline street-level image using FLUX.1 API (PiAPI, 2025). Figure 8 displays five images (from left to right): (1) a cycling lane with a different surface colour, (2) the street without cars, (3) the baseline image, (4) the cycling lane removed, and (5) an alternative pavement type. All other visual elements are held constant. These images are then evaluated with the trained CV-DCM. For each image, the figure reports its assigned utility (ranked from high to low) and, in the top-left corner, the estimated choice probability —defined as the probability of selecting the given cycling environment over the baseline for a ± 11 -minute cycling trip, all else being equal.

The colour of a cycling lane appears to matter. Changing it from grey to red increases the utility from -0.04 to 0.25 , an improvement of 0.29 . Based on the estimated travel-time coefficient ($\beta_{tt} = -0.18$), this translates into cyclists being willing to take a detour of roughly one and a half minutes longer on an 11-minute trip in order to use a red lane instead of a grey one in the same setting. In direct comparison with the baseline, about 57% of cyclists would opt for the red-lane route. Similar colour swaps in other images consistently produce higher utilities, reinforcing the indication that lane colour matters to cyclists. This aligns with previous work: Dutch operational studies link red surfacing to better safety and smoother traffic flow (Bouwgrondstoffen, 2024), while stated-preference experiments show red lanes are judged safer and more appealing than other colours (Vera-Villarreal et al., 2016). Karlsen and Fyhri (2020) add a nuance, observing that colour preference can depend on users' familiarity with coloured cycling infrastructure. Taken together, both our model results and the literature suggest that lane colour can be an effective, low-cost design intervention for improving the perceived attractiveness of cycling environments to cyclists.

The absence of cars increases the cycling environment utility by 0.16 relative to the baseline version with cars. With the estimated travel-time coefficient ($\beta_{tt} = -0.18$), this difference translates into a willingness to accept roughly 0.9 minutes of extra travel on an 11-minute trip in order to ride on a car-free road. The finding echoes the estimation results in Table 4 (Model 2), where the presence of parking, an indirect proxy for cars, decreases route utility. It is also consistent with the study of Uijtendewilligen et al. (2024), who find that traffic is a dominant factor in cycling route choices.

The utility difference between a marked lane and its absence is negligible, indicating that cyclists may be largely indifferent between these two street designs. An alternative explanation is methodological: the CV-DCM may not capture fine-scale surface markings, such as the narrow white stripe of a suggestion lane, because these details are difficult for the feature extractor to find in street-level imagery. A substantial utility difference of 0.31 emerges between asphalt and brick pavement. This aligns with the findings of Kaparias et al. (2024), who report that cyclists favour asphalt over other pavement types. It is also consistent with the disutility estimated for brick surfaces by Model 2 as shown in Table 4. Taken together, the results indicate that cyclists prefer red lanes over grey, value car-free streets, and favour smooth asphalt surfaces, while subtle lane markings offer little added appeal. It also demonstrates the CV-DCM's value as a design-assessment tool, providing planners with quantitative feedback on cycling interventions prior to making any investments.



Figure 8: Baseline image (centre) and four generated variants with their cycling environment utilities

4 Conclusion

This study investigated cyclists' preferences for different cycling environments, focusing on features such as lane type, pavement type, and greenery. To this end, we conducted a novel image-based stated route choice experiment in which respondents evaluated alternatives that varied in cycling environment, travel time, and the number of traffic lights. Street-level imagery was employed to represent the cycling environment, with more than 6,000 images used in the experiment.

We analysed the resulting choice data with both traditional discrete choice models and recently developed computer vision-enriched discrete choice models that integrate visual features extracted from images into the discrete choice modelling framework. Our findings demonstrate that the cycling environment is a significant determinant of cycling route choice behaviour. In particular, cyclists show a strong preference for separated cycling lanes and are willing to accept an additional 1.5 minutes of travel on an 11-minute trip to avoid mixed-traffic roads.

Finally, we illustrated how our models can provide planners with actionable insights. We mapped the extent to which Rotterdam's cycling network aligns—or conflicts—with cyclists' preferences and simulated targeted cycling environment design interventions to evaluate their ex-ante impact on cycling environment utility. These results highlight the potential of combining stated choice experiments with computer vision methods to support evidence-based planning for more attractive and cyclist-friendly cities.

Several limitations need to be acknowledged. While the use of real-world images in SC experiments offers significant advantages, it also raises important questions. For example, it is plausible that the visual representation of the cycling environment disproportionately captures attention, potentially diverting focus from numerically presented attributes (e.g., travel time and traffic lights in our study) and ultimately leading to biased estimates (Guilfoos et al., 2023; Li & Camerer, 2022; Murwirapachena & Dikgang, 2022). Future research could further investigate this hypothesis, e.g. through an A/B testing framework. Moreover, although images greatly enhance respondents' understanding of the choice context compared to text and numerical descriptions alone, the presented scenarios may still deviate from real-world conditions. In our case, single images are used to represent entire routes, whereas in reality, the cycling environment can vary significantly along the route. Future studies could address this limitation by incorporating a series of images or videos, perhaps even with audio, to provide a more realistic representation of the choice situation. Finally, while the CV-DCM retains interpretability for numerical attributes, the computer vision component remains opaque. Future research could address this by adopting feature-learning approaches in which semantically meaningful image features are extracted and incorporated into traditional choice models.

CRedit authorship contribution statement

Roosmarijn Terra: Writing – original draft, Conceptualisation, Software, Methodology, Formal analysis, Data curation, Visualisation. **Francisco Garrido-Valenzuela:** Conceptualisation, Methodology, Data curation, Supervision, Writing – review & editing. **Sander van Cranenburgh:** Conceptualisation, Software, Resources, Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Oded Cats:** Supervision, Funding acquisition, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the writing process

During this work the authors used ChatGPT (OpenAI) to improve grammar and clarity in the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content

of this publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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